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United States Patent	12383783
Kind Code	B2
Date of Patent	August 12, 2025
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Dual action weightlifting machine with selectorized resistance

Abstract

An exercise machine, having: (a) a frame; (b) a pair of weight stacks connected to opposite sides of the frame; (c) a pair of vertically moveable pulley carriages connected to front portions of the frame; (d) a lifting bar assembly connected to a rear portion of the frame; (e) a pair of arm cables, wherein each of the arm cables are connected to one of the pulley carriages and pass around a first series of pulleys such that the weight stacks are lifted when the arm cables are pulled by a user through the pulley carriages; and (f) a pair of lifting bar cables respectively connected to a pair of vertically moveable bar mounts of the lifting bar assembly that pass around a second series of pulleys such that the weight stacks are lifted when the lifting bar of the lifting bar assembly is raised by the user.

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Appl. No.:	18/418902
Filed:	January 22, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240157186 A1	May. 16, 2024

Related U.S. Application Data

continuation-in-part parent-doc US 18226964 20230727 US 12296217 child-doc US 18418902
us-provisional-application US 63395112 20220804

Publication Classification

Int. Cl.: **A63B21/062** (20060101); **A63B21/00** (20060101); **A63B23/035** (20060101); **A63B23/12** (20060101)

U.S. Cl.:

CPC **A63B21/0628** (20151001); **A63B21/156** (20130101); **A63B21/4035** (20151001); **A63B21/4045** (20151001); **A63B23/03525** (20130101); **A63B23/1218** (20130101);

Field of Classification Search

CPC: A63B (21/00058); A63B (21/00065); A63B (21/00069); A63B (21/00076); A63B (21/062); A63B (21/0626); A63B (21/0628); A63B (21/0724); A63B (21/078); A63B (21/0783); A63B (21/15); A63B (21/151); A63B (21/152); A63B (21/154); A63B (21/156); A63B (21/4035); A63B (21/4045); A63B (23/035); A63B (23/03508); A63B (23/03516); A63B (23/03525); A63B (23/03533); A63B (23/03541); A63B (23/03558); A63B (23/03566); A63B (23/1218); A63B (71/0054); A63B (2071/0072); A63B (2071/0081); A63B (2225/09); A63B (2225/093); A63B (2225/10); A63B (2225/105); A63B (2225/107)

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Background/Summary

RELATED APPLICATION (1) The present application is a Continuation-In-Part of U.S. patent application Ser. No. 18/226,964, of same title, filed Jul. 27, 2023, and claims priority to U.S. Provisional Patent Application Ser. No. 63/395, 112, filed Aug. 4, 2022, of same title, the full disclosures of which are incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

(1) The present invention relates to exercise machines in general and to Smith machine exercise machines of the type which have an exercise bar that is guided in two directions to simulate free-weight barbell exercise movements. The present application also relates to arm cable pull exercise machines.

SUMMARY OF THE INVENTION

(2) The present system preferably combines a guided-bar exercise machine, hereafter referred to as a “2D Smith Machine” together with a dual cable pull exercise machine, providing the advantages of both machines in a single exercise machine frame. One advantage of the present system is that it is selectorized (i.e.: it uses weight stacks as opposed to free weights) for both the 2D Smith Machine and for the two cable pull exercise assemblies. A particular advantage of the present

system is that both the cable pull exercise machine and the 2D Smith exercise machine portions of the present device share the same weight stacks (and remain connected to the weight stack regardless of which exercise the user is performing). In alternate embodiments, the front arm cable portions of the machine are omitted.

(3) Another advantage of the present system is that the lifting bar of the Smith Machine is always connected to the weight stacks. As such, it is not necessary for a user to switch a cable connector between the Smith Machine and the dual cable pull portions of the machine. This makes the present system simpler and easy to operate (as compared to exercise machines that require the user to either connect a cable to the Smith Machine portion of the exercise machine or to the dual cable arm pull portion of the machine).

(4) A further advantage of the present system is that the lifting bar of the Smith Machine need not be locked away when not in use. Instead, the lifting bar can be simply pushed to the back of the machine, with the Smith Machine's vertical supports being optionally locked into position there.

(5) In optional embodiments, the present system is provided solely as a 2D Smith machine with the dual arm cable exercise portions removed. Different cabling systems for these embodiments are illustrated.

(6) In one preferred aspect, the present pulley assemblies are configured such that the lifting bar lifts 100% of the selected weight in the weight stack(s) whereas the cable pulleys only lift 50% of the selected weight in the weight stack(s). In alternate embodiments, the lifting bar lifts 50% of the selected weight in the weight stack. It is to be understood, however, that other configurations and weight percentages are also encompassed within the scope of the present system. Specifically, different weight resistances can be provided to the lifting bar and to the cable pulleys based solely on the arrangement of the various pulleys within the present system.

(7) 2D Smith machines such as taught by Randall Webber in U.S. Pat. No. 7,713,179, have stationary racking positions for the lifting bar and adjustable “safety tiers” that insure that a lifting bar cannot accidentally fall on a user by providing a bottom height below which the lifting bar is not able to fall. Accordingly, 2D Smith machines are popular both for doing squats and for doing bench presses. Typically, a 2D Smith machine provides a frame with guard rails or latches that prevent the lifting bar from falling below the pre-set bottom height. This has the benefit of providing a more natural lifting experience. As will be shown, the present exercise device offers all of these advantages. However, with the present system, the user is able to move the lifting bar back and forth (horizontally) while also lifting the bar (vertically). This assists the user in developing balance while simultaneously developing their strength working different muscle groups.

(8) In preferred aspects, the present system comprises an exercise machine, comprising: (a) a frame; (b) a pair of weight stacks connected to opposite sides of the frame; (c) a pair of vertically moveable pulley carriages connected to front portions of the frame; (d) a lifting bar assembly connected to a rear portion of the frame, wherein the lifting bar assembly comprises: (i) a lifting bar, (ii) a vertically moveable bar mount supporting each end of the lifting bar, and (iii) a vertically moveable locking mechanism for setting a bottom height for each of the vertically moveable bar mounts; (e) a pair of arm cables, wherein each of the arm cables are connected to one of the pulley carriages and pass around a first series of pulleys such that the weight stacks are lifted when a free end of the arm cables are pulled by a user through the pulley carriages; and (f) a pair of lifting bar cables, wherein a free end of each of the lifting bar cables are connected to one of the vertically moveable bar mounts and pass around a second series of pulleys such that the weight stacks are lifted when the lifting bar is raised by the user.

(9) Preferably, the vertically moveable locking mechanism comprises: a vertically moveable lower stop on which the bar mount rests, a pull pin assembly connected to the lower stop, and a frame member having a series of vertical holes dimensioned to receive the pull pin therein. In operation, the user rotates the lifting bar to release a pull pin, thereby permitting adjustment of the vertical height of the lower stop (which sets the bottom height below which the lifting bar cannot fall).

Preferably as well, the lifting bar can be moved in a horizontal direction on the frame during a vertical lift. This provides the benefit of a more natural workout (as opposed to a more traditional Smith machine which just permits straight up and down movement of the lifting bar in a constrained, typically vertical, plane).

(10) Preferably, the vertically moveable pulley carriages are each mounted to a vertically extending carriage support that is independently rotatable about a vertical axis. The benefit of such rotation is that it permits the user to pull the arm cables from a wide variety of directions, thereby providing exercises to more of the user's muscles.

(11) In preferred aspects, the arm cables and lifting bar cables can both be passed around a transfer pulley assembly which moves in a manner such that pulling on either of the arm cable or the lift bar cable results in the weight stack being lifted to a different height. This gearing-style use of the transfer pulley assembly has the advantage of using the same weight stack for two very different exercises (in which a user typically lifts two different amounts of weight). In one specific embodiment, the placement of the transfer pulley assembly results in the user lifting half the load when performing cable arm pulls as compared to when performing lifting bar raises. It is to be understood, however, that other geometries and percentages could instead be used, all keeping within the scope of the present invention.

(12) In optional aspects where the arm cable exercise portion of the device is removed, the present system comprises an exercise machine, comprising: (a) a frame; (b) a pair of weight stacks connected to opposite sides of the frame; (c) a lifting bar assembly connected to a rear portion of the frame, the lifting bar assembly comprising: (i) a lifting bar, (ii) a vertically moveable bar mount supporting each end of the lifting bar, and (iii) a vertically moveable locking mechanism for setting a bottom height for each of the vertically moveable bar mounts; and (d) a pair of lifting bar cables, wherein each of the lifting bar cables are connected to one of the vertically moveable bar mounts and pass around a second series of pulleys such that the weight stacks are lifted when the lifting bar is raised by the user.

(13) In further optional embodiments, a pull-up assembly can be mounted onto the front of the frame and may include a pair of flip-and-grip handles. As such, the user is able to do lifting bar exercises, cable arm pull exercises and pull-up exercises on the same exercise machine frame.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front perspective view of the present 2D Smith-type exercise machine having forwardly facing dual cable columns exercise carriages.

(2) FIG. 2 is a side elevation view of the exercise machine corresponding to FIG. 1.

(3) FIG. 3 is a front elevation view of the exercise machine corresponding to FIG. 1.

(4) FIG. 4 is a rear elevation view of the exercise machine corresponding to FIG. 1.

(5) FIG. 5 is a top plan view of the exercise machine corresponding to FIG. 1.

(6) FIG. 6 is a front perspective view of the exercise machine of FIG. 1 with elements of the device removed to show operation and layout of the cables.

(7) FIG. 7 is a front perspective view of the left side of the exercise machine corresponding to FIG. 6.

(8) FIG. 8 is a side elevation view of the left side of the exercise machine corresponding to FIG. 6.

(9) FIG. 9 is a close up perspective view of the locking mechanism for the lifting bar.

(10) FIG. 10 is a view similar to FIG. 9, but taken from the other side of the device.

(11) FIG. 11 is a simplified illustration of the cabling of the exercise machine of FIGS. 1 to 10.

(12) FIG. 12 is a side elevation view of a first embodiment of the present exercise machine where the arm cable exercise portion of the machine has been removed.

- (13) FIG. 13 is a front perspective view of the exercise machine of FIG. 12.
- (14) FIG. 14 is a front perspective view of the exercise machine of FIG. 12 with elements of the device removed to show operation and layout of the cables.
- (15) FIG. 15 is a side elevation view of a second embodiment of the present exercise machine where the arm cable exercise portion of the machine has been removed.
- (16) FIG. 16 is a front perspective view of the exercise machine of FIG. 15.
- (17) FIG. 17 is a front perspective view of the exercise machine of FIG. 15 with elements of the device removed to show operation and layout of the cables.
- (18) FIG. 18 is a simplified illustration of the cabling of the exercise machine of FIGS. 15 to 17.
- (19) FIG. 19 is a simplified illustration of an alternate cabling system in which the lifting bar lifts 50% of the selected weight in the weight stack.
- (20) FIG. 20 is a perspective view of locking mechanisms for holding the lifting bars' vertical members at a rearward position.
- (21) FIG. 21 is a close-up view corresponding to FIG. 20.
- (22) FIG. 22 is a perspective view of an alternate locking mechanism for the lifting bar.

DETAILED DESCRIPTION OF THE DRAWINGS

- (23) FIGS. 1 to 5 show various views of the present combined Smith-type exercise and arm pull exercise machine. FIGS. 6 to 8 show details of the preferred cabling systems (with various system components removed for clarity of illustration). FIGS. 9 and 10 illustrate a preferred embodiment of the locking mechanism used to pre-set the lowest vertical point of travel of the lifting bar. FIG. 11 explains the cabling layout in the exercise machine of FIGS. 1 to 10. FIGS. 12 to 17 illustrate optional preferred embodiments of the present system having only a Smith-style exercise machine with no arm pull exercise machine. FIG. 18 explains the cabling layout in the exercise machine when the arm pull portion of the machine has been omitted. FIG. 19 shows an alternate cabling approach in which the lifting bar instead only lifts 50% of the selected weight in the weight stack. FIGS. 20 and 21 show locking mechanisms for holding the lifting bars' vertical members at the back of the frame. Lastly, FIG. 22 shows an alternate locking mechanism for the lifting bar.
- (24) Referring to the attached FIGS. 1 to 10, the present system provides a combined Smith-type exercise machine, cable arm pull exercise machine and pull-up exercise machine all within a single integrated frame, as follows.
- (25) Exercise machine 10 comprises a frame 20, a pair of weight stacks 30 on opposite sides of the frame 10, and a pair of vertically moveable pulley carriages 40 mounted to the front of frame 10 as shown. Also included is a lifting bar assembly 50 comprising a lifting bar 52, a vertically moveable bar mount 54 supporting each end of lifting bar 52, and a vertically moveable locking mechanism 56 for setting a bottom height of travel for each of the vertically moveable bar mounts 54. Also included are a pair of arm cables 60, wherein each of the arm cables are connected to one of the pulley carriages 40 and pass around a first series of pulleys 70 such that the weight stacks 30 are lifted when the free end of arm cables 40 are pulled by a user through the pulley carriages 40. In addition, a pair of lifting bar cables 80 are also provided. A free end of each of the lifting bar cables 80 are connected to one of the vertically moveable bar mounts 54 and pass around a second series of pulleys 90 such that the weight stacks 30 are lifted when lifting bar 52 is raised by the user.
- (26) In preferred aspects, the vertically moveable locking mechanism 56 comprises: a vertically moveable lower stop 57 on which bar mount 54 rests, a pull pin assembly 58 connected to lower stop 57, and a frame member 59 having a series of vertical holes dimensioned to receive the pull pin therein. In operation, rotation of lifting bar 52 releases a pull pin 53 in the pull pin assembly 58, thereby permitting vertical adjustment of the height of the lower stop. This may be accomplished by a flange 51 on lifting bar 52 engaging with pull pin assembly 58 when the lifting bar 52 is rotated.
- (27) As can also be seen, one end 81 of the lifting bar cables 80 is connected to one of the bar mounts 54. Another free end 83 is connected to lower stop 57. As such, when the user raises lifting

bar **52**, the free end **81** of cable **80** is raised, and cable **80** is thereby tightened, in turn lowering transfer pulley assembly **100**. This in turn has the effect of pulling on cable **60** which in turn raises weight stack **30**. As can be seen, one of each of arm cables **60** and lifting bar cables **80** are disposed on either side of the frame. In the present system, no cables are required to pass from one side of the frame to the other. This has the beneficial effect of keeping the center space of frame **10** open for the user to perform exercises in.

(28) Preferably, the vertically moveable pulley carriages **40** are each mounted to a vertically extending carriage support **42** that is rotatable about a vertical axis. This permits the user to pull on cables **60** from a wide variety of angles, thereby exercising different parts of the user's arms and back.

(29) In preferred aspects, the lifting bar **52** and lifting bar assemblies **50** are moveable in a horizontal direction on the frame. This is best seen in FIGS. **1**, **5** and **9** where lifting bar **52** and lifting bar assemblies **50** are supported on a vertical member **110** which together with vertical member **111** is moveable back and forth in a horizontal direction along horizontal member **120**. As seen in FIGS. **20** and **21**, the vertical member **111** can be fixed into position at the rear of the machine by pins **112**.

(30) One of each of the arm cables **60** and the lifting bar cables **80** pass around transfer pulley assemblies **100**, as shown. In preferred aspects, the positioning of pulleys **70** and **90** and transfer pulley assemblies **100** is configured such that the vertical height of transfer pulley assemblies **100** remains constant when the arm cable **60** is pulled to raise the weight stack **30**, but the vertical height of the transfer pulley assemblies **100** changes when the lifting bar cable **80** is pulled to raise weight stack **30**. As a result, pulling the arm cable **60** and the lifting bar cable **80** an equal distance results in weight stack **30** moving different distances. For example, pulling the arm cable **60** a distance of X inches lifts the weight stack a distance $\frac{1}{2} X$ inches, whereas pulling the lifting arm cable **80** the same distance of X inches lifts the weight stack by distance X inches.

(31) FIG. **11** is a simplified illustration of the cabling of the exercise machine of FIGS. **1** to **10**.

(32) In further preferred aspects, a pull-up assembly **150** is mounted to the front of frame **10**. Pull-up assembly **150** may optionally comprise a pair of flip-and-grip handles **152**.

(33) FIGS. **12** to **17** show embodiments of the present exercise machine, but with the optional cable arm exerciser portion removed. FIGS. **12** to **14** show a first embodiment having the option to add a cable arm exerciser thereto (by later replacing the cables shown with longer ones), and FIGS. **15** to **17** show a second embodiment in which a cable arm exerciser cannot be added thereto. FIGS. **15** to **17** illustrate the simplicity in cabling the 2D Smith-type portion of the exercise device in this particular embodiment. FIG. **18** is a simplified illustration of the cabling of the exercise machine of FIGS. **15** to **17**. FIG. **19** is an alternate cabling system similar to FIG. **18**, but in which the lifting bar instead lifts only 50% of the selected weight in the weight stack. FIGS. **20** and **21** show how the vertical support for the lifting bar can be locked into position out of the way at the back of the machine when not in use.

(34) Lastly, FIG. **22** shows an alternate locking mechanism **50B** for the lifting bar **52**. In this alternate embodiment, slidable assembly **54B** supports the ends of lifting bar **52**. Slidable assembly **54B** may have three rollers as illustrated to engage opposite sides of vertical member **110B** such that slidable assembly **54B** can travel up and down along vertical member **110B**. Rotation of lifting bar **52** moves hook **51B** which in turn retracts biased pull pin **53B**. Pull pin **53** engages holes **53C** in vertical member **110B** such that moveable lower stop **57B** can then be set to a desired height.

Claims

1. An exercise machine including a combined Smith Machine and cable pull exercise machine, the exercise machine comprising: a frame; a weight stack mounted on the frame; a lifting bar with a lifting bar cable connected thereto, wherein the lifting bar is connected to a rear portion of the

frame; an arm cable, wherein a free end of the arm cable is located at a front portion of the frame; and a transfer pulley assembly, wherein both the lifting bar cable and the arm cable pass around the transfer pulley assembly; wherein raising the lifting bar causes the lifting bar cable to raise one or more weights on the weight stack, and wherein pulling on the arm cable causes the arm cable to raise one or more weights on the weight stack.

2. The exercise machine of claim 1, wherein the lifting bar and the arm cable are independently operable such that movement of either raises the one or more weights on the weight stack.
3. The exercise machine of claim 1, wherein both the lifting bar cable and the arm cable remain in contact with the transfer pulley assembly during either a lifting bar exercise or an arm pull exercise.
4. The exercise machine of claim 1, wherein the arm cable passes around a first series of pulleys to lift the one or more weights on the weight stack, and wherein the lifting bar cable passes around a second series of pulleys to lift the one or more weights on the weight stack.
5. The exercise machine of claim 1, wherein the lifting bar travels along a vertical member, and wherein the lifting bar is positionable in a horizontal direction.
6. The exercise machine of claim 5, wherein the lifting bar can be locked into position at the rear portion of the frame.
7. The exercise machine of claim 1, wherein the vertical height of the transfer pulley assembly remains constant when the arm cable is pulled to raise the weight stack.
8. The exercise machine of claim 1, wherein the vertical height of the transfer pulley assembly changes when the lifting bar cable is pulled to raise the weight stack.
9. The exercise machine of claim 1, wherein pulling the arm cable and the lifting bar cable an equal distance results in the weight stack moving different distances.
10. The exercise machine of claim 1, further comprising: a vertically movable bar mount supporting an end of the lifting bar, and a vertically movable locking mechanism for setting a bottom height for the vertically movable bar mount.
11. The exercise machine of claim 10, wherein the vertically moveable locking mechanism comprises: a vertically moveable lower stop on which the vertically movable bar mount rests; a pull pin assembly connected to the vertically movable lower stop; and a frame member having a series of vertical holes dimensioned to receive a pull pin of the pull pin assembly therein.
12. The exercise machine of claim 11, wherein rotation of the lifting bar releases the pull pin of the pull pin assembly, thereby permitting adjustment of a vertical height of the vertically movable lower stop.
13. The exercise machine of claim 1, further comprising: a vertically movable pulley carriage mounted to the front of the frame.
14. The exercise machine of claim 13, wherein the vertically movable pulley carriage is rotatable around a vertical axis.
15. The exercise machine of claim 1, wherein a height of the lifting bar can be adjusted without raising the one or more weights on the weight stack.
16. The exercise machine of claim 1, further comprising: a pull-up assembly mounted to the front of the frame.
